

True or Artefactual Hyperprolactinemia: Should Patients Be Systematically Screened for Macroprolactin?

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Background & Aims

Macroprolactinemia is a benign form of hyperprolactinemia caused by high-molecular-weight prolactin complexes (macroprolactin). Because it can be mistaken for true hyperprolactinemia, failure to identify it may result in misdiagnosis and unnecessary treatment. Whether screening for macroprolactin should be systematic or restricted to patients selected by a prolactin threshold remains debated.

This study aimed to assess the prevalence of macroprolactinemia among hyperprolactinemic patients and to compare the diagnostic yield of systematic screening with selective strategies based on two literature-derived thresholds (≥ 30 and ≥ 50 ng/mL).

Methods

A cross-sectional study was conducted at the CPMC hormone laboratory between December 2024 and June 2025.

Among 566 patients screened, 116 hyperprolactinemic samples (under 200 ng/mL) were included :

- in males : >15.2 ng/mL,
- in females : >23.3 ng/mL

Serum prolactin concentrations were measured using the Elecsys® Prolactin II immunoassay (Roche Diagnostics), and PEG precipitation was performed on all samples.

Recovery (R%) was calculated as : $R\% = (\text{post-PEG PRL} / \text{pre-PEG PRL}) \times 100$

Macroprolactinemia was assessed using two complementary criteria: (1) a recovery (R%) $<40\%$ and (2) a post-PEG monomeric prolactin within the assay-specific reference interval.

Results

Macroprolactinemia prevalence was 5.2% (95% CI, 2.4–10.8%) using the recovery criterion and 6.9% (95% CI, 3.5–13.0%) using the post-PEG monomeric prolactin criterion (Table 1).

Table 1: Prolactin concentrations of subjects before and after PEG precipitation

Characteristics	All patients (N=116)	Macroprolactinemia (N=8) (6.9%)	True hyperprolactinemia (N=108) (93.1%)	p value
Age (mean $\pm$ SD)	33.6 $\pm$ 10.9	36.8 $\pm$ 15.3	33.4 $\pm$ 10.5	0.399
Sex F/M, N (%)	95(81.9)/21(18.1)	7(87.5)/1(12.5)	88(81.5)/20(18.5)	1.000
Total (pre-PEG) prolactin (median, IQR), ng/mL	38.5 (30.6–55.9)	30.9 (25.1–40.9)	38.7 (31.3–58.4)	0.062
Post-PEG prolactin (median, IQR), ng/mL	27.4 (22.3–40.0)	12.1 (9.4–14.4)	28.3 (23.2–40.3)	<0.001
% Recovery (median, IQR)	72.9 (62.5–81.4)	30.2 (23.7–51.5)	74.3 (65.2–81.6)	<0.0001

Table 2: Distribution of patients according to recovery and monomeric prolactin

Recovery category	n (%)	Monomeric PRL within the reference interval, n (%)	Elevated monomeric PRL, n (%)
$<40\%$	6 (5.2)	5 (83.3)	1 (16.7)
40–60% (gray zone)	18 (15.5)	3 (16.7)	15 (83.3)
$>60\%$	92 (79.3)	0 (0.0)	92 (100.0)
Total	116 (100)	8 (6.9)	108 (93.1)

Discussion

Restricting screening to samples with prolactin concentrations ≥ 30 ng/mL still required PEG precipitation in 77.6% of patients while identifying 50% of macroprolactinemia cases (Table 3). Increasing the threshold to ≥ 50 ng/mL reduced testing to 31.0% of samples but detected only 12.5% of cases (Figure 1). Increasing the screening threshold substantially reduced diagnostic yield, providing only a modest reduction in laboratory workload at the lower threshold and an unacceptable loss of sensitivity at the higher threshold.

Table 3: Performance of screening strategies

Screening strategy	Patients tested n(%)	True Positive	False Negative	Sensitivity, % [IC95 %]	p value
Systematic screening	116 (100)	8	0	100,0 [67,6–100,0]	–
PRL ≥ 30 ng/mL	90 (77,6)	4	4	50,0 [21,5–78,5]	0,125
PRL ≥ 50 ng/mL	36 (31,0)	1	7	12,5 [2,2–47,1]	0,016

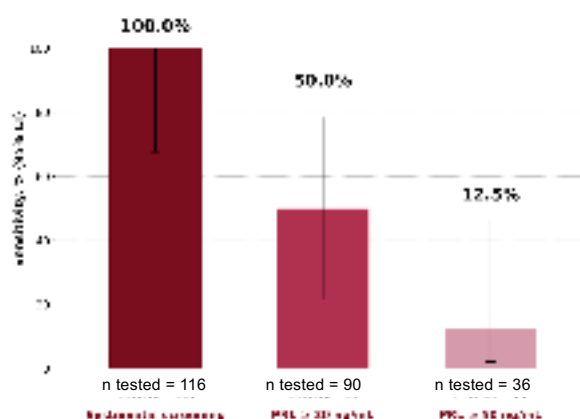


Figure 1: Detection sensitivity of macroprolactinemia according to screening strategies

Conclusion

These findings support systematic macroprolactin screening in hyperprolactinemic patients rather than threshold-based testing, which failed to reduce laboratory workload without compromising case detection.

Reporting post-PEG monomeric prolactin concentrations, their assay-specific reference intervals, and recovery percentages would improve the interpretation of hyperprolactinemia.